

XRPL Validator — Daily Engineering Report • 2026-07-20

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **6 commits** (xrpl-validator 6)

Engineering changes

- **fix(xrpl-ledger): issuer TickSize in offer placement — corpus 95.0%** (a5db381 , xrpl-validator) OfferCreate ignored the issuer-published TickSize. rippled rounds the requested rate UP to N significant digits (Quality::round) and re-derives the non-exact side — TakerPays for tfSell, TakerGets otherwise — BEFORE crossing, so both the crossing and the placed remainder use the adjusted amounts. Mainnet tx 9870DA80 asked to sell 8539920 drops for STX (issuer TickSize 6) and rippled stored 8539914 with no ...
- **fix(xrpl-ledger): route strand output to the destination — corpus 95.6%** (bb1eeef , xrpl-validator) A cross-currency Payment credited the acquired asset to the SENDER and then forwarded it to the destination. rippled delivers the strand's output straight to the destination: routing through the sender materializes an intermediate trust line it never creates — and when the destination IS the issuer (mainnet tx AAA6EB389D3A: RIBBLE paid to rG7jT6D4, its own issuer), the IOU is redeemed at the maker/AMM line rather ...
- **fix(xrpl-ledger): trust-line lifecycle — receiver-only reserve, NoRipple, zero-balance deletion — corpus 95.6%** (d547205 , xrpl-validator) Three faithful-port fixes in line_adjust, from rippled rippleCreditIOU / trustCreate: - trustCreate charges the reserve to the RECEIVER only. We bumped BOTH parties' OwnerCount, so the issuer accrued phantom reserves. The line still joins both owner directories; only one OwnerCount moves. - New lines get NoRipple on the receiver's side unless their account has DefaultRipple. This flag is also part of the deletion ...
- **feat(xrpl-ledger): tfSell crossing semantics — sell the whole TakerGets — corpus 96.3%** (3a3f337 , xrpl-validator) A tfSell offer sells its ENTIRE TakerGets even if that acquires more than TakerPays; TakerPays is only a minimum. Our crossing engine capped each fill at rem_pays (the wanted amount) and terminated the walk once it was reached, so a tfSell that could fully sell only by taking the surplus was left with an unsold remainder — and a FillOrKill/ImmediateOrCancel sell was then wrongly killed. Mainnet tx 2042F8CE / ...
- **fix(xrpl-ledger): sell-side offer funding check — tecUNFUNDED_OFFER — corpus 96.8%** (870eefd , xrpl-validator) An OfferCreate can only sell what the account actually holds of the TakerGets asset. Our crossing engine checked the MAKER funding but never the taker: after the tfSell fix started crossing the full TakerGets, five arb offers from one account selling IOUs it holds NONE of (CLARITY, XALPHA) crossed and succeeded, where mainnet returns tecUNFUNDED_OFFER fee-only. Verified: the account CLARITY line was absent at seq-1.
- **fix(xrpl-ledger): faithful OfferCreate preclaim funding — tecUNFUNDED_OFFER — corpus 96.8%** (5b1d386 , xrpl-validator) OfferCreate::preclaim returned the generic TxResult::Unfunded (tecUNFUNDED) and

rejected whenever `balance < the FULL sell amount`. Both are wrong: rippled unfunds only when `accountFunds(TakerGets) <= 0` (for XRP: balance minus reserve — partially funded offers still cross) and the code is `tecUNFUNDED_OFFER`. The old check was also XRP-only.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **On-Ledger Positive Validator Reputation Score** (consensus) Add a singleton `ValidatorReputation` ledger object (keylet analogous to `NegativeUNL`'s, reserve-free, capped to current UNL size) populated once per f
- **Time-Limited NFT Lease Via Session Keys** (nft) Add a new `NFTTokenLease` transaction that creates an owner-reserved `ltNFTOKEN_LEASE` object (keylet derived from `NFTTokenID` + sequence) binding an exi
- **Revocable Hash-Commitment Attestation Registry** (audit) Add a new `ltATTESTATION` ledger object (owner-reserved by the issuer, `keylet::attestation(issuer, sequence)`) created via `AttestationSet` , storing

Appendix — commits

Time	Repo	Commit	Subject
10:24	xrpl-validator	a5db381	fix(xrpl-ledger): issuer TickSize in offer placement — corpus 95.0%
12:04	xrpl-validator	bb1eefa	fix(xrpl-ledger): route strand output to the destination — corpus 95.6%
16:28	xrpl-validator	d547205	fix(xrpl-ledger): trust-line lifecycle — receiver-only reserve, NoRipple, zero-balance deletion — corpus 95.6%
18:19	xrpl-validator	3a3f337	feat(xrpl-ledger): tfSell crossing semantics — sell the whole TakerGets — corpus 96.3%
19:49	xrpl-validator	870eefd	fix(xrpl-ledger): sell-side offer funding check — tecUNFUNDED_OFFER — corpus 96.8%
21:19	xrpl-validator	5b1d386	fix(xrpl-ledger): faithful OfferCreate preclaim funding — tecUNFUNDED_OFFER — corpus 96.8%